

Physics 7645 - Introduction to the Standard Model

Spring 2026

Lectures	TH	2:55-4:10 PM	RCK B15	Anders Ryd
Backup	F	8:40-9:55 AM	RCK B15	Anders Ryd

Contact: 385 PSB 5-2529 Anders.Ryd@cornell.edu

- Content:

This course is an introduction to the standard model of particle physics. Familiarity with Feynman rules, Lagrangians, and relativistic wave equations at the level of PHYS 7651 is required. Topics covered include strong and electro-weak interactions, Higgs mechanism, and phenomenology of weak interactions, the quark model, and particle detectors. The course is taught at the level of *The Standard Model and Beyond* by Paul Langacker.

- Prerequisites

Equivalent of PHYS 7651, QFT1, are required. If you have not taken this classes at Cornell come speak to me.

- Books:

<i>Standard Model and Beyond</i>	Langacker	<i>Required</i>
<i>The Standard Model: a Primer</i>	Burgess and Moore	
<i>Electroweak Interactions</i>	Renton	
<i>Particle Physics: A comprehensive Introduction</i>	Seiden	
<i>Elementary Particle Physics in a Nutshell</i>	Tully	
<i>Quarks and Leptons</i>	Halzen and Martin	

There are many other books on the topic. You should consider buying one (or many) of them.

- CANVAS:

You should be automatically enrolled in the CANVAS course for Physics 7645. I will be sending out announcements via the linked email account so make sure to check it regularly. Homework will also be distributed via CANVAS.

- Course Material, in broad strokes:

- Overview of the standard model
- Review of Dirac Equation, Spinors, Group Theory
- Gauge theories
- Quantum Electrodynamics as a Gauge Theory
- Quantum Chromodynamics
- Electroweak theory, from V-A to Higgs Mechanism
- Experiments

- Homework and Grade Policy

In addition to weekly homework, there will be a final project, to be completed by the end of the term with a paper. Your final grade will be composed of a *participation grade (10%)*, a *homework grade (65%)*, and a *project grade (25%)*.

For the participation grade, this is a small class! Engage with the material and with your fellow students and me. We will all have more fun that way.

Homework will be posted on Canvas and will be due a week later, in class. If more than three days late it will not be accepted.

For the homework, I encourage you to work in groups; however, the final homework you turn in should be your own and list with whom you worked.

For the project we will work together to define the project topics. Expect this to happen towards the end of March.

- Academic Integrity

Cornell takes issues of academic integrity extremely seriously. Integrity and honesty will be absolutely essential in your future careers. Students who are found in violation of Cornell's Code of Academic Integrity can expect to be prosecuted to the full extent allowed by the code. Some examples of violations for which we have prosecuted students in the recent past:

- Turning in homework that has been copied from another student or downloaded from the internet. Working in groups is encouraged; however, your write-up must be your own.
- Copying the work of another student during a quiz or exam.
- Using cheat sheets, notes written on body appendages, clothing and hats, or programmed calculators during a quiz or exam.
- Changing an answer on a quiz or exam after it has been graded and submitting it for a regrade.
- Providing false reasons (e.g., illness, death in the family, interviews, and athletic events) for missing a quiz or exam. Proper documentation must be provided.
- Providing false reasons in requesting a make-up exam. Proper documentation must be provided.
- Copying material from sources such as Wikipedia without proper attribution or citations. Working with references is crucial in science so you should be familiar with it.

No matter how desperate your situation may seem, you should never violate the code of academic integrity and never make misrepresentations to your TAs or your professors. Come talk to your me, your TA, or your advising dean for advice on how to resolve your difficulties.

See the [Cornell Code of Academic Integrity](#) online for more information.

- TURNITIN.

Students agree that by taking this course all required papers may be subject to submission for textual similarity review to [turnitin.com](#) for the detection of plagiarism. All submitted papers will be included as source documents in the [turnitin.com](#) reference database solely for the purpose of detecting plagiarism of such papers. Use of the [turnitin.com](#) service is subject to the Usage Policy posted on the [turnitin.com](#) site.

- Commercial use of course materials.

Course materials posted on Canvas are intellectual property belonging to the author. Students are not permitted to buy or sell any course materials without the express permission of the instructor. Such unauthorized behavior constitutes academic misconduct.

Tentative Syllabus

Physics 7645

Spring 2026

Lect	Date	Topic	Reading
1	Jan 20	Class Intro, SM pheno recap	Lecture Notes
2	Jan 22	SM recap continued	and Ch. 1
3	Jan 27	SM recap; Conventions; QFT recap	Ch. 2
4	Jan 29	Fermionic Lagrangians	Ch. 2
5	Feb 3	Gauge Symmetries	Ch. 3.1
6	Feb 4	Group Theory 101	Ch. 3.1
7	Feb 10	Global symmetries in QFT	Ch. 3.2
8	Feb 12	SU(3) flavor - quark model	Ch. 3.2
–	Feb 17	WINTER BREAK	–
9	Feb 19	Discrete symmetries (C,P,T)	Ch. 3.2
10	Feb 25	Non-Abelian gauge theories	Ch. 4.1 and 4.2
11	Feb 26	QCD	Ch. 5.1 - 5.3
12	Mar 3	Parton model; DIS	Ch. 5.4 and 5.5
13	Mar 5	Weak interaction & Fermi theory	Ch. 6
14	Mar 10	GWS: electroweak theory	Ch. 6
15	Mar 12	GIM Mechanism; CKM	
16	Mar 17	Mass terms in QFTs	Review Ch. 2
17	Mar 19	Higgs mechanism: real scalar fields	Ch. 7.2 and 7.3
18	Mar 24	Higgs mechanism: complex scalar fields	Ch. 7.2 and 7.3
19	Mar 26	Quark masses in the SM	Ch. 7.6
–	Mar 31	SPRING BREAK	–
–	Apr 2	SPRING BREAK	–
20	Apr 7	CP violation	Ch. 7.6
21	Apr 9	ν s and ν mass	Ch. 7.7
22	Apr 14	ν oscillations	Ch. 7.7
23	Apr 16	Experiment 1	–
24	Apr 21	Experiment 2	–
25	Apr 23	Experiment 3	–
26	Apr 28	Recap and summary	–
27	Apr 30	Recap and summary	–
28	May 5	STUDENT PROJECTS PRESENTATIONS	–